

**14. Ans: A**

For the maxima to coincide, the angle of diffraction are the same for both the red and blue light.

Since  $d \sin \theta = n\lambda$ , and  $\theta$  is the same,

$$n_{red} \lambda_{red} = n_{blue} \lambda_{blue}$$

$$\frac{n_{red}}{n_{blue}} = \frac{\lambda_{blue}}{\lambda_{red}} = \frac{411}{685} = \frac{3}{5}$$

Hence 3<sup>rd</sup> order of red will coincide with 5<sup>th</sup> order of blue.